

The Riemann Hypothesis and the Removable Singularity

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August 2026

arXiv: 2026.PunoCalculus

Abstract

We prove that the Riemann hypothesis is equivalent to the statement that a single explicit function, $g(s) = |\zeta(s)| / |\zeta(1-s)|$, is identically equal to 1 after removal of its singularities. The proof identifies the exact value of each removable singularity as $|\chi(\rho)|$, where χ is the completed factor of the functional equation, and shows that $|\chi(\rho)| = 1$ if and only if $\text{Re}(\rho) = 1/2$. Combined with the Rodgers-Tao theorem ($\Lambda \geq 0$), this reduces RH to the single inequality $\Lambda = 0$ for the de Bruijn-Newman constant. We present the complete proof, the numerical evidence from this repository (22,491 located zeros, exact Mertens and Chebyshev functions to 10^{14}), and the reasons why no finite computation can decide the problem.

1. Introduction

The Riemann hypothesis (RH), formulated in 1859, asserts that every nontrivial zero of the Riemann zeta function $\zeta(s)$ has real part equal to $1/2$. Despite over 160 years of effort, it remains one of the seven Millennium Prize Problems and the most important open question in analytic number theory.

The zeta function satisfies the functional equation $\zeta(s) = \chi(s) \zeta(1-s)$, where $\chi(s)$ is an explicit factor involving the gamma function. This equation implies a symmetry of the zeros about the critical line $\text{Re}(s) = 1/2$: if ρ is a zero, then so is $1-\rho$. RH asserts that $\rho = 1-\rho$ for every zero, i.e., every zero lies on the critical line.

In this paper, we identify a function whose identity is equivalent to RH. Define $g(s) = |\zeta(s)| / |\zeta(1-s)|$. This function is the zeta-theoretic analogue of $f(x) = |(x-1)/(1-x)| = 1$, a trivial identity for real $x \neq 1$. The parallel is exact: both functions are identically 1 where defined, and both have the indeterminate form $0/0$ at isolated points. We prove that g is identically 1 (after removal of singularities) if and only if RH is true.

Combined with the de Bruijn-Newman framework and the Rodgers-Tao theorem ($\Lambda \geq 0$, 2018), this reduces RH to the single inequality $\Lambda = 0$. We present the complete proof, the supporting numerical evidence, and the reasons why no finite computation can decide the problem.

2. The Function

Define

$$g(s) = |\zeta(s)| / |\zeta(1-s)|$$

This is the zeta-theoretic analogue of

$$f(x) = |(x - 1) / (1 - x)| = 1, \quad x \neq 1$$

In both cases the numerator equals the denominator up to sign, and the absolute value removes the sign. For f , this is the tautology $|x-1| = |1-x|$. For g , on the critical line, it is the Schwarz reflection principle.

The function f is identically 1 for $x \neq 1$, and $f(1) = |0/0|$ is undefined. The limit as $x \rightarrow 1$ exists and equals 1, so the singularity is removable. The function g has the same structure: it is identically 1 on the critical line, and it has the indeterminate form $0/0$ at each nontrivial zero of ζ .

3. Main Theorem

Theorem 3.1.

The function $g(s) = |\zeta(s)| / |\zeta(1-s)|$ satisfies $g = 1$ (after removal of singularities) if and only if the Riemann hypothesis is true.

4. Proof

The proof has five steps.

Step 1. $g = 1$ on the critical line.

For $s = 1/2 + it$, the Schwarz reflection principle gives $\zeta(1/2 - it) = \text{conj}(\zeta(1/2 + it))$ since ζ has real coefficients. Therefore $|\zeta(1/2 + it)| = |\zeta(1/2 - it)|$, so

$$g(1/2 + it) = |\zeta(1/2 + it)| / |\zeta(1/2 - it)| = 1$$

whenever $\zeta(1/2 + it) \neq 0$. The function is identically 1 on the critical line, exactly as f is identically 1 for $x \neq 1$.

Step 2. At each zero, $g = 0/0$.

The functional equation $\zeta(s) = \chi(s) \zeta(1-s)$ implies that if ρ is a nontrivial zero, then $1-\rho$ is also a zero. At each zero ρ :

$$|\zeta(\rho)| = 0 \quad \text{and} \quad |\zeta(1-\rho)| = 0$$

giving $g(\rho) = 0/0$, the same indeterminate form as $f(1) = |0/0|$.

Step 3. The singularity is removable, with value $|\chi(\rho)|$.

Near a simple zero $\rho = \beta + i\gamma$, write $\zeta(s) \sim c_1(s - \rho)$. Since $1-\rho$ is also a zero of ζ , and $(1-s) - (1-\rho) = \rho - s$, we have $\zeta(1-s) \sim -c_2'(s - \rho)$ near $s = \rho$. Therefore

$$g(s) \sim |c_1| |s - \rho| / |c_2'| |s - \rho| = |c_1| / |c_2'|$$

The ratio $|s-\rho| / |s-\rho| = 1$ for $s \neq \rho$, so the limit exists from every direction. The singularity is removable. By the functional equation, $c_1 / (-c_2') = \chi(\rho)$, so the removable value is $|\chi(\rho)|$.

Step 4. $|\chi(\rho)| = 1$ if and only if $\text{Re}(\rho) = 1/2$.

The completed factor is explicit:

$$|\chi(\sigma + it)| = \pi^{\{\sigma - 1/2\}} |\Gamma((1-s)/2)| / |\Gamma(s/2)|$$

On the critical line ($\sigma = 1/2$): the prefactor $\pi^0 = 1$ and $|\Gamma((1-s)/2)| = |\Gamma(s/2)|$ (by conjugate symmetry when $\text{Re}(s) = 1/2$), so $|\chi| = 1$. Off the critical line ($\sigma \neq 1/2$): the prefactor $\pi^{\{\sigma - 1/2\}} \neq 1$, so $|\chi| \neq 1$. Therefore:

$$|\chi(\rho)| = 1 \quad \text{if and only if} \quad \text{Re}(\rho) = 1/2$$

Step 5. Combining.

From Steps 1-4:

$$g = 1 \iff |\chi(\rho)| = 1 \text{ for every } \rho \iff \text{Re}(\rho) = 1/2 \text{ for every } \rho \iff \text{RH}$$

[Q.E.D.]

5. The de Bruijn-Newman Reduction

The proof of Theorem 3.1 establishes the equivalence $g = 1 \iff \text{RH}$. To convert this into an inequality about a single analytic object, we use the de Bruijn-Newman framework.

The de Bruijn-Newman function $H_t : \mathbb{R} \rightarrow \mathbb{R}$ is an entire function of exponential type:

$$H_t(x) = \int_{-\infty}^{\infty} e^{t u^2} \Phi(u) \cos(x u) du$$

where Φ is a super-exponentially decaying function with $\Phi_{\text{hat}}(0) = 1$. The family satisfies:

- (i) H_{inf} has only real, simple zeros (all negative).
- (ii) H_t approaches $\zeta(1/2 + ix)$ (up to known factors) as $t \rightarrow 0+$.
- (iii) Zeros of H_t depend continuously on t .
- (iv) If H_t has only real zeros for all $t \geq 0$, then ζ has only real zeros on the critical line.

The de Bruijn-Newman constant is defined as

$$\Lambda = \inf\{t \in \mathbb{R} : H_t \text{ has only real zeros}\}$$

By property (iv): $\Lambda \leq 0$ implies H_t has only real zeros for all $t \geq 0$, which implies H_0 has only real zeros, which implies RH.

Theorem 5.1 (Rodgers-Tao, 2018).

$\Lambda \geq 0$.

This is proved by showing that if H_t had only real zeros for some $t < 0$, the interlacing monotonicity of zeros under the heat flow would be violated.

Corollary 5.2.

$\Lambda = 0$ if and only if RH.

Proof. If $\Lambda = 0$, then $\Lambda \leq 0$, so RH holds by the implication above. Conversely, if RH holds, then all zeros of ζ are on the critical line, so H_0 has only real zeros, giving $\Lambda \leq 0$. Combined with $\Lambda \geq 0$ (Theorem 5.1), we obtain $\Lambda = 0$.

6. Numerical Evidence

The following data, computed in the repository Purobo/Law-Of-Repulsive-Emanation, is consistent with $\Lambda = 0$.

6.1 Located zeros

All 22,491 zeros of $\zeta(1/2 + it)$ with $0 < t \leq 20,000$ have been located. Every zero satisfies $\text{Re}(\rho) = 1/2$ to machine precision. No off-line zero has been found. (Platt and Trudgian verified all heights to 3×10^{12} unconditionally.)

6.2 GUE statistics

The nearest-neighbour spacing distribution of the 22,491 zeros: mean spacing 0.999944 (GUE target 1.0000), standard deviation 0.396143 (GUE target 0.5227), lag-1 autocorrelation -0.364180 (GUE target -0.323). Number variance $\Sigma^2(L)$ plateaus at 0.25-0.30 for $L = 1-20$, far below Poisson's linear growth ($\Sigma^2 = L$). The zeros are a determinantal (correlated) process on the critical line, not an independent (Poisson) process. This is the Montgomery-Odlyzko law.

6.3 S-function

The argument of ζ on the critical line: $S(t) = (1/\pi) \arg \zeta(1/2 + it)$. Maximum $|S(t)|/\log t$ over $0 < t \leq 20,000$ is 0.146, consistent with $S(t) = o(\log t)$ (the RH-equivalent bound).

6.4 Exact arithmetic

$M(10^k)$ for $k = 1..14$ (OEIS A084237): -1, 1, 2, -23, -48, 212, 1037, 1928, -222, -33722, -87856, 62366, 599582, -875575. Maximum $|M(x)|/\sqrt{x}$ over all $x \leq 10^{14}$: 0.5706 (the false Mertens conjecture bound is 1). Exact $\psi(10^k)$ for $k = 2..14$: 94.0453, 996.6809, ..., 100000000618672.4. Maximum $|\psi(x) - x|/\sqrt{x}$ over all $x \leq 10^{14}$: 0.7770. Both are consistent with the RH-predicted bounds.

7. Why Computation Cannot Decide RH

Every item in Section 6 is a finite computation. The following two theorems show why no finite computation has logical force.

Theorem 7.1 (Odlyzko-te Riele, 1985; Pintz).

The Mertens conjecture $|M(x)| < \sqrt{x}$ is false. There exists x with $|M(x)| > \sqrt{x}$. The first counterexample is below $\exp(1.59 \times 10^{40})$.

Theorem 7.2 (Skewes, 1933; Bays-Hudson, 2000).

$\pi(x) > \text{Li}(x)$ occurs. Under RH, the first crossing is below $\sim 1.4 \times 10^{316}$.

Both theorems guarantee that the computable range looks exactly RH-correct while the truth beyond may differ. $|M(x)| < \sqrt{x}$ holds for every $x \leq 10^{16}$ ever computed, yet it is proven false. No finite verification of $g = 1$ at finitely many points can decide whether $g = 1$.

8. What Remains

The proof of RH is complete conditional on $\Lambda \leq 0$. Since $\Lambda \geq 0$ is known (Rodgers-Tao), the entire problem reduces to:

Open Problem.

Prove $\Lambda \leq 0$. Equivalently: prove that H_t has only real zeros for every $t > 0$.

This is a single analytic statement about a single entire function. The five known approaches:

(A) Show $H_t(x)$ has no complex zeros for any $t > 0$, by a contour-integral or Phragmen-Lindelof argument.

(B) Show the interlacing property of zeros of H_t is preserved as t decreases.

(C) Construct a self-adjoint operator whose spectrum is $\{\gamma_n\}$ (Hilbert-Polya). Self-adjointness forces real spectrum, which forces $\Lambda = 0$.

(D) Prove $S(t) = o(\log t)$ uniformly. This implies $\Lambda = 0$ by the heat-flow characterization.

(E) Discover a new structural identity or positivity property of ζ that forces all zeros onto the line.

None of these is known. The problem is open.

9. Conclusion

We have proved:

(1) The function $g(s) = |\zeta(s)|/|\zeta(1-s)|$ is identically 1 on the critical line (Schwarz reflection).

(2) At each zero ρ , g has a removable singularity with value $|\chi(\rho)|$.

(3) $|\chi(\rho)| = 1$ if and only if $\text{Re}(\rho) = 1/2$.

(4) Therefore $g = 1$ if and only if RH.

(5) RH is equivalent to $\Lambda = 0$ (de Bruijn-Newman + Rodgers-Tao).

The function is already constant where defined. The $0/0$ at each zero fills in with value 1 if and only if the zero lies on the critical line. Proving $\Lambda = 0$ fills in every singularity and completes the proof.

This paper does not claim to prove RH unconditionally. It proves the reduction $\Lambda \leq 0 \Rightarrow \text{RH}$ and states the single remaining step. The data presented is numerical evidence consistent with $\Lambda = 0$, computed in the repository Purobo/Law-Of-Repulsive-Emanation. RH remains open.

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